

ADAM HAMILTON (He/Him/His)

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EDUCATION

Brandeis University, Waltham, MA

Fall 2024–May 2025

MS, Neuroscience

GPA: 4.0

- Coursework focused on developmental cognition, neuropsychological assessment, and statistical modeling
- Courses: Developmental Cognition, Childhood Disorders, Advanced Psych Stats
- Capstone Project: *Neuromelanin-MRI of Substantia Nigra and Executive Function in Aging*
Advisor: Dr. Anne Berry

University of New Hampshire, Durham, NH

Spring 2021

BS, Neuroscience and Behavior

GPA: 3.6 | Honors

RESEARCH EXPERIENCE

The Neurochemistry and Cognition Lab

Fall 2024–May 2025

Graduate Researcher

Principal Investigator: Dr. Anne Berry

- Led development of a neuromelanin-sensitive MRI processing pipeline to quantify structural integrity of the substantia nigra in aging populations
- Assessed patient executive function through neuropsychological tasks
- Assisted study protocols under IRB guidance
- Utilized multi-level modeling statistical approaches and PET techniques to investigate dopaminergic system's impact on aging pathologies and cognitive processes
- Collaborated on data tracking, QC, and reporting for large behavioral and imaging datasets

Champlain College Neuroscience Laboratory, Burlington, VT

Summer 2022-2023

Research Assistant and Co-author

Colombo, B., Hamilton, A., Telazzi, I., Balzarotti, S. (2023). The relationship between cognitive reserve and the spontaneous use of emotion regulation strategies in older adults: a cross-sectional study. *Aging Clinical and Experimental Research*, 35(7), 1505-1512.

- Co-Authored published peer-reviewed article on cognitive reserve and emotional regulation in aging adults
- Conducted literature reviews and assisted with IRB submissions for study rationale
- Surveyed participants using emotional regulation questionnaire (ERQ) and cognitive reserve test (Core T)
- Administered behavioral and emotional regulatory protocols to young adults
- Operated tDCS equipment and tracked participant responses
- Managed scoring and data storage

SKILLS

Behavioral Assessments and Neuropsychological tasks: Stroop, N-back, Keep-Track, Core-T

Data analysis & Programming: R, MATLAB, Excel, Python, Bash

Neuroimaging: MRI and PET imaging analysis methods

Language: Proficiency in Spanish

Certifications: CITI Certified for Human Subjects Research, Responsible Conduct of Research, Social and Behavioral Research